

SAFE-TRIAGE API DOCUMENTATION

OVERVIEW

SAFE-Triage provides clinical decision support for emergency department triage using a hybrid AI + deterministic rules approach.

Base URL (Cloud Run):

- <https://safe-triage-eciux5h4aq-uc.a.run.app>

ENDPOINTS

POST /AI-TRIAGE

Description: AI-assisted triage assessment (Gemini + UMLS + deterministic ESI rules).

Request

```
{
  "patient_id": "string",
  "age": 58,
  "gender": "male",
  "chief_complaint_text": "chest pain radiating to arm",
  "vitals": {
    "hr": 95,
    "sbp": 140,
    "dbp": 90,
    "rr": 18,
    "temp": 37,
    "spo2": 97
  },
  "consciousness": "A",
  "comorbidities": ["diabetes"]
}
```

Response (example)

```
{
  "level": 2,
  "ai_data": {
    "snomed_code": "225566008",
    "category": "chest_pain_cardiac",
    "icd10_coding": {"primary_code": "I20.9"}
  },
  "red_flags": [],
}
```

```
"reasoning": [...],
"icd10_coding": {"primary_code": "I20.9"}
}
```

POST /TRIAGE

Description: Deterministic triage (no AI).

POST /CONFIRM-TRIAGE

Description: Record human confirmation/override of triage decision.

Request

```
{
  "patient_id": "string",
  "recommended_esi": 2,
  "confirmed_esi": 2,
  "clinician_id": "nurse_01",
  "clinician_role": "nurse",
  "action": "confirmed",
  "override_reason": null,
  "supervisor_pin": null
}
```

POST /CONFIRM-TRIAGE/REQUEST

Description: Register pending confirmation (for timeout/queue).

GET /PENDING-CONFIRMATIONS

Description: List pending confirmations.

POST /TRANSCRIBE

Description: Speech-to-text (ar-EG) using Google Speech-to-Text.

Multipart Form

- 'audio': audio file (webm/ogg/wav)
- 'language': 'ar-EG' (default)

GET /HEALTH

Description: Health check.

ERROR CODES

- '400': validation error
- '401': invalid supervisor PIN
- '403': unauthorized downgrade
- '500': internal error

SAFE-TRIAGE DEPLOYMENT GUIDE

PREREQUISITES

- Google Cloud project with billing enabled
- UMLS API key
- Firebase project for frontend hosting

ENVIRONMENT VARIABLES

- 'UMLS_API_KEY': UMLS API key
- 'USE_VERTEX_SPEECH': 'true' to enable Speech-to-Text
- 'SUPERVISOR_PIN': Supervisor PIN for downgrades (default '0000')
- 'CONFIRMATION_TIMEOUT_SECONDS': default '300'

BACKEND DEPLOY (CLOUD RUN)

```
gcloud run deploy safe-triage \
  --source ./backend \
  --region us-central1 \
  --memory 2Gi \
  --cpu 2 \
  --min-instances 1 \
  --max-instances 10 \
  --allow-unauthenticated \
  --project safe-triage-ai \
  --set-env-vars="UMLS_API_KEY=...
,UUSE_VERTEX_SPEECH=true,SUPERVISOR_PIN=0000,CONFIRMATION_TIMEOUT_SECONDS=300"
```

FRONTEND DEPLOY (FIREBASE)

```
cd frontend
npm run build
firebase deploy --only hosting
```

SPEECH-TO-TEXT ENABLEMENT

```
gcloud services enable speech.googleapis.com --project safe-triage-ai
gcloud projects add-iam-policy-binding safe-triage-ai \
  --member="serviceAccount:459364571026-compute@developer.gserviceaccount.com" \
  --role="roles/speech.client"
```

MEDGEMMA QA JOB

```
gcloud run jobs deploy medgemma-qa-review \
  --source ./backend \
  --region us-central1 \
  --project safe-triage-ai \
  --command python \
  --args jobs/medgemma_qa_job.py
```

```
gcloud scheduler jobs create http medgemma-qa-schedule \
  --location us-central1 \
  --schedule "*/15 * * * *" \
  --uri "https://us-central1-run.googleapis.com/apis/run.googleapis.com/v1/namespaces/safe-triage-ai/jobs/medgemma-qa-review:run" \
  --http-method POST \
  --oauth-service-account-email 459364571026-compute@developer.gserviceaccount.com
```

SAFE-TRIAGE USER GUIDE

FOR TRIAGE NURSES

1. Enter patient demographics
2. Record chief complaint (voice or text)
3. Enter vital signs
4. Click ****Assess Patient****
5. Review recommendation and confirm ESI within 5 minutes

CONFIRMING ESI

- Click ****Confirm ESI**** to accept AI recommendation
- If overriding, select new ESI and provide a reason
- Downgrades require supervisor PIN

FOR SUPERVISORS

- Review pending confirmations
- Enter PIN to approve downgrades
- Monitor QA flags (MedGemma job)

TROUBLESHOOTING

- If voice input fails, use manual text entry
- If offline, system uses cached SNOMED + ICD-10
- If confirmation times out, case escalates to supervisor

PERFORMANCE BENCHMARK REPORT

METHOD

- Warm instance ('min-instances=1' enabled)
- Attempted repeated POST requests to '/ai-triage'

RESULTS (THIS ENVIRONMENT)

- '/health' response time: ~0.42s
- Single successful '/ai-triage' request: ~2.39s
- Repeated POST benchmarking is unreliable due to intermittent DNS resolution in this environment.

RECOMMENDATION

- Re-run 'backend/performance_benchmark.py' from local machine or Cloud Shell for full metrics.
- Target: mean < 0.5s, P95 < 0.8s, P99 < 1.5s

COLD START

- Improved from 17.4s to ~0.45s (97% faster)

SAFE-TRIAGE THESIS DEFENSE SLIDES (OUTLINE)

1. Title: SAFE-Triage â AI-Powered Emergency Triage for Egyptian Hospitals
2. Problem: ED overcrowding, unreliable internet, language barriers
3. Research Question & Objectives

4. System Architecture (3-layer model)
5. Hybrid Approach: AI extracts, rules decide, human confirms
6. UMLS RAG + SNOMED-CT integration
7. Offline Mode (6,370 SNOMED + ICD-10 cache)
8. Arabic Support (ar-EG speech + bilingual UI)
9. Human Confirmation Workflow
10. MedGemma QA (async review)
11. Alerts (Telegram â Email)
12. Validation: 138 scenarios, zero under-triage
13. Performance: cold start 0.45s
14. Case Study: Chest pain â ESI 2 (SNOMED 225566008, ICD-10 I20.9)
15. Limitations (over-triage)
16. GAHAR Compliance
17. Future Work
18. Conclusion
19. Q&A
